```sql
-- Create Database
CREATE DATABASE company_db;

USE company_db;

-- Create Employee Table
CREATE TABLE employee(
    id INT PRIMARY KEY AUTO_INCREMENT,
    name VARCHAR(50),
    age INT,
    department VARCHAR(50),
    salary DECIMAL(10,2)
);

-- Insert Records
INSERT INTO employee(name,age,department,salary)
VALUES
('Rahul',22,'IT',45000),
('Aman',23,'HR',40000),
('Neha',24,'Sales',55000),
('Priya',25,'IT',60000);

-- Display All Records
SELECT * FROM employee;

-- WHERE Clause
SELECT * FROM employee
WHERE department='IT';

-- ORDER BY
SELECT * FROM employee
ORDER BY salary DESC;

-- DISTINCT
SELECT DISTINCT department
FROM employee;

-- UPDATE Record
UPDATE employee
SET salary=50000
WHERE id=1;

-- DELETE Record
DELETE FROM employee
WHERE id=2;

-- Aggregate Functions
SELECT COUNT(*) FROM employee;

SELECT MAX(salary) FROM employee;
```

```sql
SELECT MIN(salary) FROM employee;

SELECT AVG(salary) FROM employee;

SELECT SUM(salary) FROM employee;


-- Create Department Table
CREATE TABLE department(
    dept_id INT PRIMARY KEY AUTO_INCREMENT,
    dept_name VARCHAR(50)
);

-- Insert Records
INSERT INTO department(dept_name)
VALUES
('IT'),
('HR'),
('Sales');

-- Add Department Id in Employee Table
ALTER TABLE employee
ADD dept_id INT;

UPDATE employee SET dept_id=1 WHERE id=1;
UPDATE employee SET dept_id=2 WHERE id=3;
UPDATE employee SET dept_id=3 WHERE id=4;

-- Add Foreign Key
ALTER TABLE employee
ADD CONSTRAINT fk_department
FOREIGN KEY(dept_id)
REFERENCES department(dept_id);

-- INNER JOIN
SELECT e.name,d.dept_name
FROM employee e
INNER JOIN department d
ON e.dept_id=d.dept_id;

-- LEFT JOIN
SELECT e.name,d.dept_name
FROM employee e
LEFT JOIN department d
ON e.dept_id=d.dept_id;

-- Show employee names only
SELECT name
FROM employee;
```

```sql
-- Show employee name and salary
SELECT name, salary
FROM employee;

-- Show employees with age greater than 22
SELECT *
FROM employee
WHERE age > 22;

-- Show employees with salary less than 60000
SELECT *
FROM employee
WHERE salary < 60000;

-- Show employees working in IT department
SELECT *
FROM employee
WHERE department = 'IT';

-- Show employees whose salary is between 45000 and 60000
SELECT *
FROM employee
WHERE salary BETWEEN 45000 AND 60000;

-- Show employees working in IT or Sales
SELECT *
FROM employee
WHERE department IN ('IT','Sales');

-- Show employee names starting with R
SELECT *
FROM employee
WHERE name LIKE 'R%';

-- Show employee names ending with a
SELECT *
FROM employee
WHERE name LIKE '%a';

-- Show employee names containing h
SELECT *
FROM employee
WHERE name LIKE '%h%';

-- Display unique departments
SELECT DISTINCT department
FROM employee;

-- Display employees in ascending order of age
```

```sql
SELECT *
FROM employee
ORDER BY age ASC;

-- Display employees in descending order of salary
SELECT *
FROM employee
ORDER BY salary DESC;

-- Count total employees
SELECT COUNT(*)
FROM employee;

-- Find maximum salary
SELECT MAX(salary)
FROM employee;

-- Find minimum salary
SELECT MIN(salary)
FROM employee;

-- Find average salary
SELECT AVG(salary)
FROM employee;

-- Find total salary
SELECT SUM(salary)
FROM employee;

-- Count employees department wise
SELECT department, COUNT(*)
FROM employee
GROUP BY department;

-- Average salary department wise
SELECT department, AVG(salary)
FROM employee
GROUP BY department;

-- Departments having more than one employee
SELECT department, COUNT(*)
FROM employee
GROUP BY department
HAVING COUNT(*) > 1;

-- Employee with highest salary
SELECT *
FROM employee
WHERE salary = (
    SELECT MAX(salary)
```

```
    FROM employee
);

-- Employee with lowest salary
SELECT *
FROM employee
WHERE salary = (
    SELECT MIN(salary)
    FROM employee
);

-- Show all departments
SELECT *
FROM department;

-- Display department names only
SELECT dept_name
FROM department;
```